

## Lab 7

**NO** late submission. Grade = 0 after the deadline.

**Due:** 13:50h 19/11/2025

Submission:

- **Lab#\_YOURNAME.ipynb**
- Must include all data file (**.csv**,...) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the **same** folder (`pd.read_csv('data.csv')`)
- No use of LOCAL libraries, NO local directory for libraries

Use '**recipe\_missing.csv**' (missing dataset) and '**recipe.csv**' (original dataset)

- a) The dataset has missing data. Show the percentage of missing data for each feature/column in '**recipe\_missing.csv**' and then plot in a bar chart.
- b) Use **KNNImputer**,  $k = 5$ . Plot the histogram of imputed data and original data of the last 3 columns. Compute the Mean Square Error (MSE) between original data and imputed data.

```
from sklearn.impute import KNNImputer
imputer = KNNImputer(n_neighbors=3)
```

- c) Use **MICE** algorithm for **10** cycles. Initialize missing data using **SimpleImputer**, using "mean" strategy. For each cycle, fill in the missing place using **IterativeImputer** and estimator=**DecisionTree** Regressor. Plot the histogram of imputed data and original data of the last 3 columns. Compute the Mean Square Error (MSE) between original data and imputed data.

```
from sklearn.impute import IterativeImputer
from sklearn.tree import DecisionTreeRegressor

simple_imputer = SimpleImputer(strategy='mean')
...
```

```
imputer = IterativeImputer(estimator=DecisionTreeRegressor(),  
max_iter=10, random_state=0)  
  
imputed_data = imputer.fit_transform(data)
```

e. Compare the methods in c) and d)

**Notice:**

Mean Squared Error (MSE) measures the average of the squared differences between predicted (imputed) values and actual values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- $y_i$  = actual (true) value
- $\hat{y}_i$  = predicted value
- $n$  = number of observations